

# Deep Learning Project Proposal

*Design and Creative Technologies*

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# Review Pulse v2: Aspect-Based Sentiment Analysis of Customer Reviews with Attention-Based Deep Learning

## 1. Abstract

Customer reviews often praise one aspect while criticising another, yet sentence-level sentiment systems, including our Assessment 1 N-gram classifier, reduce a review to one label. This proposal designs **Review Pulse v2**, an aspect-based sentiment analysis (ABSA) system that predicts sentiment for specified aspects. Four models are compared on SemEval-2014 Restaurants: TF-IDF, a target-agnostic LSTM, ATAE-LSTM and DistilBERT. The study evaluates performance, efficiency and indicative token-level evidence, then packages aspect-aware predictions in a feasible Streamlit prototype for Assessment 3.

## 2. Problem Statement, Aim and Research Questions

**Problem.** Most sentiment systems assign one polarity to an entire text. Our Assessment 1 N-gram classifier and the deep CNN of Zhao, Gui and Zhang (2018) do this. However, "*the food was great but the service was slow*" contains opposing opinions about two aspects. A single label hides the detail that product, hospitality and customer-experience teams need to act on.

**Aim.** Design and evaluate a low-compute ABSA system that predicts sentiment for a specified aspect, measures the value of aspect conditioning, compares ATAE-LSTM with DistilBERT, and presents indicative token-level evidence where the model supports attention or attribution.

### Research questions.

- **RQ1** - How much does explicit aspect conditioning improve sentiment classification on multi-aspect sentences compared with a target-agnostic baseline?
- **RQ2** - How do ATAE-LSTM and DistilBERT compare on accuracy, macro-F1 and computational efficiency?
- **RQ3** - What human-readable evidence do attention or attribution visualisations provide for aspect-level predictions?

## 3. Literature Review

Deep learning changed sentiment analysis by learning representations rather than relying only on engineered features. Zhao, Gui and Zhang (2018) report that a convolutional network over word vectors outperforms traditional baselines for Twitter sentiment. Nevertheless, their model, like Review Pulse v1, produces one label per text. Pontiki et al. (2014) define SemEval-2014 Task 4 and its aspect-level annotations, shifting the practical question from whether a review is positive to which aspect is positive.

Sequence models address this gap through target conditioning. Tang et al. (2016) show that target-dependent LSTMs improve over target-agnostic variants by representing context around the target, although they do not selectively weight every context token. Wang et al. (2016) add an aspect embedding and attention in ATAE-LSTM, allowing the same sentence to be represented differently for each aspect. This makes ATAE-LSTM a suitable lightweight model and exposes attention weights, but the representation is still learned from a comparatively small benchmark and may overfit.

Transformers offer a different trade-off. Devlin et al. (2019) introduce pretrained bidirectional contextual representations, while Sun, Huang and Qiu (2019) reformulate ABSA as sentence-pair classification using an auxiliary aspect sentence. DistilBERT applies this contextual advantage with lower compute than full BERT, but it remains more expensive than ATAE-LSTM and neither transformer attention nor post-hoc attribution is automatically a faithful causal explanation. Recent transformer benchmarks (Jayakody et al., 2024), zero- and few-shot LLM studies (Simmering & Huoviala, 2023), and the review by Hua et al. (2024) position these models within a rapidly expanding field.

Accordingly, the project does not assume that the Transformer will win. It tests whether pretrained context justifies its additional cost against ATAE-LSTM's lighter, aspect-conditioned design. The comparison therefore combines predictive performance, efficiency and indicative interpretability evidence. Neural aspect discovery (He et al., 2017) remains a possible extension, not a requirement for the core experiment.

## 4. Proposed Approach and Methods

**Dataset and aspect origin.** SemEval-2014 Task 4 Restaurants is the primary domain; Laptops is an optional cross-domain extension. Training and evaluation use the dataset's gold aspect terms. The original *conflict* polarity label is removed from the three-class positive/neutral/negative task and its exclusions are counted and reported. In Streamlit v2, a user instead enters a review and manually specifies one or more aspects. Automatic aspect extraction and Topic Modelling remain outside the core scope.

**Model progression.** TF-IDF with logistic regression and a target-agnostic LSTM consume only the review and establish sentence-level baselines. ATAE-LSTM consumes *(review, aspect)* through an aspect embedding and aspect-aware attention. DistilBERT also consumes *(review, aspect)*, representing the aspect as an auxiliary sentence. This four-model progression isolates recurrence, explicit conditioning and pretrained contextual representation.

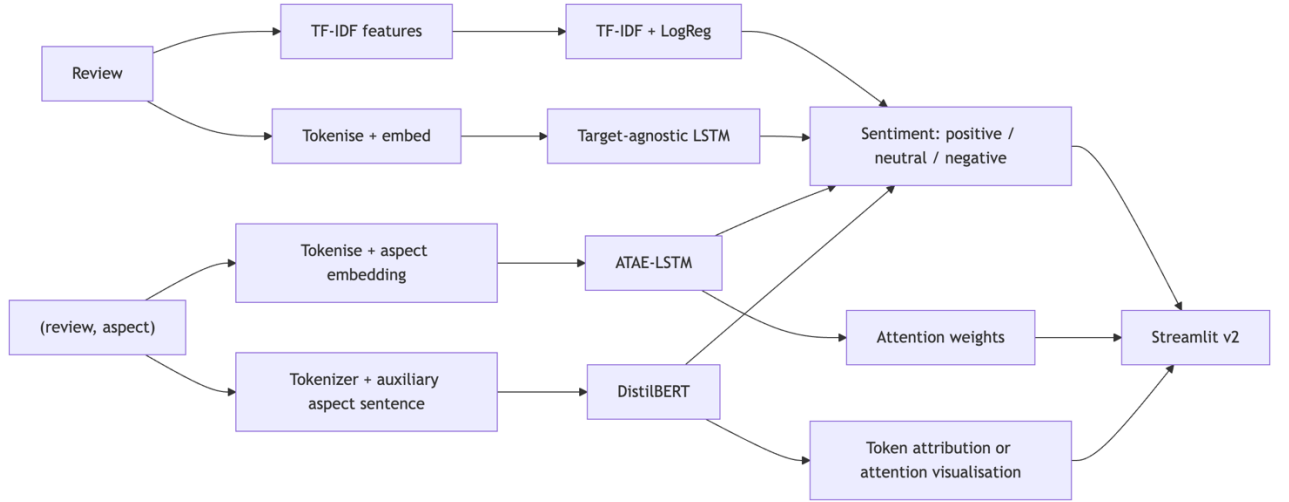


Figure 1. Review Pulse v2 pipeline. The sentence-level baselines receive only the review; the aspect-conditioned models receive a *(review, aspect)* pair. Only attention- or attribution-capable models produce token-level visual evidence.

**Evaluation.** Accuracy, macro-F1, class-level results and efficiency are reported on fixed train/development/test splits. Splitting is grouped by sentence before aspect instances are formed, preventing the same review from leaking across partitions. A **mixed-polarity multi-aspect subset** contains sentences with at least two aspects associated with different polarities; it tests the cases that one sentence-level label cannot resolve.

Table 1. Assessment 3 reporting shell. The last column evaluates sentences with at least two aspects carrying different polarities.

| Model                                                  | Accuracy            | Macro-F1            | Accuracy on mixed-polarity multi-aspect subset |
|--------------------------------------------------------|---------------------|---------------------|------------------------------------------------|
| <b>TF-IDF + logistic regression (review only)</b>      | to be reported (A3) | to be reported (A3) | to be reported (A3)                            |
| <b>Target-agnostic LSTM (review only)</b>              | to be reported (A3) | to be reported (A3) | to be reported (A3)                            |
| <b>ATAE-LSTM (review, aspect; attention)</b>           | to be reported (A3) | to be reported (A3) | to be reported (A3)                            |
| <b>DistilBERT (review, aspect; auxiliary sentence)</b> | to be reported (A3) | to be reported (A3) | to be reported (A3)                            |

**Interpretability output.** For predictions from attention- or attribution-capable models, the interface presents the aspect label and indicative token-level evidence. ATAE-LSTM exposes attention weights; DistilBERT uses token attribution or an attention visualisation. Table 2 is an illustrative expected interface output, not an experimental result. TF-IDF and the target-agnostic LSTM are not required to produce heatmaps.

Table 2. Illustrative expected output for one multi-aspect review. Bold text stands in for token shading. Attention is indicative evidence, not a causal explanation unless faithfulness tests support that interpretation.

Review: "The food was great but the service was slow."

| Aspect  | Predicted sentiment | Indicative token-level evidence (bold = highest weight)             |
|---------|---------------------|---------------------------------------------------------------------|
| food    | positive            | the · <b>food</b> · was · <b>great</b> · but · service · was · slow |
| service | negative            | the · food · was · great · but · <b>service</b> · was · <b>slow</b> |

**Deployment.** The Assessment 3 prototype lets a user enter a review plus one or more aspects, then returns a sentiment for each aspect and model-supported visual evidence.

## 5. Project Plan and Risk Management

Work is divided into dated, reviewable outcomes (Table 3). Luis owns the technical foundation; group ownership marks integration points requiring peer review and shared acceptance. A fixed seed, versioned data audit, saved configurations and a single evaluation script support reproducibility. The group will review milestone evidence at each hand-off and reduce scope before moving a due date. Appendix B records probability, impact, mitigation and a concrete contingency for technical and collaboration risks.

*Table 3. Delivery plan from data audit to Assessment 3 package.*

| Aspect                 | Date           | Outcome                             | Owner  |
|------------------------|----------------|-------------------------------------|--------|
| Data + baseline        | 18-26 Jul      | Audited dataset and TF-IDF baseline | Luis   |
| LSTM + ATAE-LSTM       | 27 Jul – 2 Aug | Checkpoint and metrics              | Luis   |
| DistilBERT             | 3-8 Aug        | Trained transformer                 | Victor |
| Evaluation + interface | 9-13 Aug       | Comparison and Streamlit v2         | Juan   |
| Report + package       | 14-19 Aug      | Report, code and ZIP                | Group  |

The critical path is dataset audit, aspect-safe splitting, model training, common evaluation, interface integration and packaging. Zero-cost compute constrains Transformer experiments, so the accepted minimum product remains the audited baselines, ATAE-LSTM, shared evaluation and working interface. DistilBERT is retained when compute and validation checks pass; optional Laptops transfer, automatic aspect extraction and Topic Modelling are cut first.

## 6. Conclusion

Sentence-level sentiment loses actionable aspect detail. Review Pulse v2 provides a controlled four-model comparison, with Restaurants as the core domain, explicit aspect-safe evaluation and a dated delivery plan. The prototype will visualise indicative token-level evidence where supported, without presenting attention as model reasoning or a causal explanation.

## 7. Appendices

### 7.1. Appendix A – From Assessment 1 to Review Pulse v2

Assessment 1 established a transparent sentence-level baseline using observed N-gram counts. Assessment 2 proposes the transition from fixed local probabilities to learned contextual representations and from one sentiment label per text to one label per specified aspect. The evaluation discipline remains continuous across both assessments.

*Table A1 – Knowledge transition from Assessment 1 to the proposed Assessment 2 and 3 project.*

| Assessment 1 - N-gram sentiment                 | Assessment 2/3 - Review Pulse v2                                                |
|-------------------------------------------------|---------------------------------------------------------------------------------|
| Count observed word sequences                   | Learn distributed and contextual representations                                |
| Fixed Markov context window                     | LSTM recurrence, aspect-aware attention and Transformer context                 |
| One sentiment label per tweet                   | One sentiment label per aspect within a review                                  |
| Add-k smoothing controls sparse counts          | Dropout, weight decay, early stopping and transfer learning control overfitting |
| Hand-defined probability and threshold rule     | Learned logits optimized through loss and backpropagation                       |
| Inspect bigram probabilities and error examples | Inspect attention or attribution alongside error examples                       |
| Accuracy, macro-F1 and confusion matrices       | Retain the same metrics, now calculated for aspect-level predictions            |

Review Pulse v1 also supplies reusable preprocessing, experiment, metric and interface patterns. The Assessment 3 implementation changes the model input from review-only baselines to explicit (*review, aspect*) pairs for ATAE-LSTM and DistilBERT, then exposes per-aspect predictions in Streamlit v2.

## 7.2 Appendix B – Risk Register

Table B1 – Risk Register and Mitigation plan.

| <b>Risk</b>                                                    | <b>Probability</b> | <b>Impact</b> | <b>Mitigation</b>                                                              | <b>Contingency</b>                                                         |
|----------------------------------------------------------------|--------------------|---------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Overfitting on the small benchmark</b>                      | High               | High          | Stratified development checks, dropout, early stopping and fixed seeds         | Reduce model capacity and report variance across seeds                     |
| <b>Transformer compute exceeds the free tier</b>               | Medium             | High          | DistilBERT, short pilot runs, capped sequence length and checkpointing         | Ship ATAE-LSTM as the primary deep model and report the compute constraint |
| <b>Leakage between aspects from the same sentence</b>          | Medium             | High          | Split by sentence identifier before expanding aspect instances; audit overlaps | Rebuild all splits and rerun every affected result                         |
| <b>Artifact or model loading failure</b>                       | Medium             | High          | Pin dependencies, save configs and test a clean-start load before packaging    | Bundle a verified lightweight checkpoint and fall back to CPU inference    |
| <b>Scope creep from Laptops, extraction or Topic Modelling</b> | High               | Medium        | Gate all extensions until the core acceptance tests pass                       | Remove extensions in that order and preserve the Restaurants core          |
| <b>Unequal group contribution</b>                              | Medium             | High          | Named owners, dated evidence, twice-weekly reviews and a contribution log      | Reassign overdue work within 24 hours and document the revised ownership   |

End of Appendix Section

## Statement of Acknowledgment

We acknowledge that we have used the following AI tools in the preparation of this project proposal:

- OpenAI ChatGPT (Codex-5.5)
- Anthropic Claude (Opus 4.8)

Both tools were used to assist with clarifying aspect-based sentiment analysis (ABSA) concepts and the progression from N-gram baselines to attention-LSTM and transformer models, structuring the literature review and its critical synthesis, sanity-checking the SemEval-2014 experimental design against data leakage, refining academic clarity, and supporting APA 7th referencing conventions. All source papers were read and verified against the original publications by the group.

Prompt examples:

1. *"Given our Assessment 1 N-gram single-label classifier, how do I frame the aspect-level problem so that a sentence like 'the food was great but the service was slow' motivates ABSA over sentence-level sentiment?"*
2. *"Compare ATAE-LSTM (Wang et al., 2016) and BERT-for-ABSA via auxiliary sentence (Sun et al., 2019) as two contrasting models on SemEval-2014 Task 4, and explain what each adds over a target-agnostic baseline."*
3. *"Review this project plan for a hypothetical ABSA proposal: are the risks, contingency plan and critical success factors sound and feasible within a fixed academic timeframe and a zero-cost compute budget?"*

We confirm that the use of these AI tools has been in accordance with the Torrens University Australia Academic Integrity Policy and the TUA, Think and MDS Position Paper on the Use of AI. We confirm that the final output is authored by the group and represents our own critical thinking, analysis, and synthesis of sources. We take full responsibility for the final content of this report.

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